SunCast: Learning Solar Flare Forecasting

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Abstract

Accurate prediction of solar flares is critical for mitigating the impact of space weather on satellites and communication systems. Existing machine learning approaches often model solar activity as independent or weakly coupled time series, overlooking interactions between solar active regions. We introduce SunCast, a spatio-temporal framework that represents solar active regions as nodes in a dynamically learned interaction graph and jointly models spatial dependencies and temporal evolution. SunCast employs a graph attention–based architecture to capture adaptive inter-region interactions, followed by a recurrent temporal module that learns short- and long-term dynamics for flare forecasting. Experiments on a real-world multi-class solar flare dataset show that SunCast achieves an 8% improvement in accuracy over non-graph baselines, with consistent performance across multiple random initializations. These results highlight the importance of explicitly modeling spatial interactions for reliable solar flare forecasting.

1 Introduction

Solar flares are explosive releases of magnetic energy in the solar corona that emit intense Extreme Ultra-Violet (EUV), X-ray, and gamma-ray radiation. Severe flaring events threaten modern technological infrastructure, causing radiation hazards for astronauts, disruptions to satellite navigation and communication systems, and damage to ground-based electronics. The economic impact of extreme space weather events is estimated to reach trillions of dollars, motivating accurate solar flare forecasting.

Predicting solar flares remains difficult due to the lack of a direct theoretical link between photospheric magnetic field evolution and flare occurrence in solar active regions (ARs). As a result, current forecasting approaches rely on data-driven models trained on historical and real-time observations. The Helioseismic Magnetic Imager (HMI) onboard the Solar Dynamics Observatory provides rich spatio-temporal measurements of magnetic field parameters describing AR evolution. Effective prediction therefore requires models that capture both temporal dynamics and inter-variable interactions.

Recent work formulates flare prediction as a multivariate time series (MVTS) classification problem, where instances represent temporal evolutions of magnetic field parameters over an observation window and are labeled by flare activity in a subsequent prediction window. MVTS-based methods outperform single-timestamp approaches by leveraging temporal context. However, existing models either use handcrafted statistical features or apply recurrent neural networks (RNNs) and long short-term memory (LSTM) architectures that primarily model temporal dependencies while ignoring inter-variable relationships.

Solar magnetic activity is inherently spatio-temporal, governed by coupled physical processes. Neglecting inter-variable interactions limits the expressive power of purely temporal models. To address this, we propose **SunCast**, a spatio-temporal deep learning framework that represents MVTS

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instances as sequences of dynamically evolving graphs, where nodes correspond to magnetic field variables and edges encode their statistical dependencies.

Our contributions are:

1. A spatio-temporal MVTS framework for solar flare prediction that explicitly models inter-variable dependencies via attention-based graph learning.
2. A dynamic graph construction strategy with Graph Attention Networks to adaptively learn variable importance over time.
3. An integration of graph-based spatial representations with LSTM-based temporal modeling to capture local and global temporal patterns.
4. Significant performance improvements over strong MVTS-based baselines on a benchmark solar flare prediction dataset.

The **SunCast** implementation and evaluation code are publicly available at `https://github.com/AI-Club-IITM/GAT-LSTM`.

2 Related Work

Early efforts in solar flare prediction relied on expert-driven systems rather than data-driven modeling. McIntosh [1990] proposed one of the earliest operational flare prediction systems, THEO, which relied on manual input of sunspot classifications and magnetic field properties and was later adopted by the Space Environment Center (SEC) of the National Oceanic and Atmospheric Administration (NOAA). While effective at incorporating domain knowledge, such systems lacked scalability and adaptability to the increasing volume and complexity of solar observations.

With the increasing availability of large-scale magnetic field measurements from space missions, solar flare prediction has shifted toward data-driven and machine learning approaches. Methods are broadly classified by the magnetic field data used into line-of-sight and vector magnetogram–based models. Early work relied on line-of-sight measurements, while the Solar Dynamics Observatory (SDO) enabled near-continuous full-disk vector magnetograms through the Helioseismic Magnetic Imager (HMI) [Mason and Hoeksema, 2010]. As a result, most recent models exploit vector magnetic field data for its richer physical information content.

Initial data-driven approaches focused on linear statistical modeling to identify magnetic field features correlated with flare productivity. Several studies reported statistically significant relationships between photospheric magnetic field parameters and flare occurrence using correlation analysis and linear discriminant analysis [Cui et al., 2006, Jing et al., 2006, Leka and Barnes, 2003]. However, such linear models are limited in their ability to capture the nonlinear and complex dynamics inherent in solar magnetic activity.

To address these limitations, nonlinear machine learning methods have been explored, including logistic regression [Song et al., 2009], decision tree–based approaches [Yu et al., 2009], support vector machines [Bobra and Couvidat, 2015, Nishizuka et al., 2017], relevance vector machines [Al-Ghraibah et al., 2015], and shallow neural networks [Ahmed et al., 2013]. More recently, convolutional neural networks (CNNs) have been applied directly to magnetogram images for automated spatial feature extraction [Park et al., 2018, Li et al., 2020, Zheng et al., 2019, Nishizuka et al., 2021]. However, CNN-based models typically operate on single timestamps or short temporal contexts, limiting their ability to model long-term temporal evolution.

Temporal modeling was introduced by framing solar flare prediction as a multivariate time series (MVTS) classification problem. Angryk et al. [2020] released a benchmark MVTS dataset in which each instance captures the temporal evolution of magnetic field parameters over a fixed observation window and is labeled according to flare activity in a subsequent prediction window. Subsequent work explored statistical summarization of univariate time series with traditional classifiers [Hamdi et al., 2017], decision tree–based MVTS models [Ma et al., 2017], and end-to-end sequence learning using recurrent neural networks such as LSTMs [Muzaheed et al., 2021]. Although LSTM-based models automate temporal feature learning, they primarily model temporal dependencies while treating variables independently, thereby neglecting structured inter-variable relationships.

Recent advances in multivariate time series analysis emphasize the importance of modeling dependencies among variables using graph-based representations. Graph neural networks (GNNs) have been successfully applied to MVTS forecasting by representing variables as nodes and their relationships as edges, enabling joint learning of spatial and temporal patterns [Wu et al., 2020, Ruiz et al., 2021]. However, the application of graph-based deep learning methods to solar flare prediction remains relatively limited, particularly with respect to dynamically modeling inter-variable interactions.

In contrast to prior work, the proposed approach explicitly models both temporal dynamics and inter-variable relationships in MVTS data for solar flare prediction. By representing magnetic field parameters as nodes in dynamically constructed graphs and leveraging attention-based graph neural networks, the model adaptively learns the relative importance of variable interactions over time. This spatial modeling is integrated with temporal learning to more effectively reflect the underlying physical complexity of solar magnetic activity.

3 Methodology

3.1 Problem Formulation and MVTS Representation

We model solar flare prediction as a multivariate time series (MVTS) classification problem. Each solar active region is represented as an MVTS instance $S \in \mathbb{R}^{T \times N}$, where T denotes the number of timestamps in the observation window and N denotes the number of magnetic field parameters. Each instance is associated with a class label indicating the flare intensity observed within a subsequent prediction window.

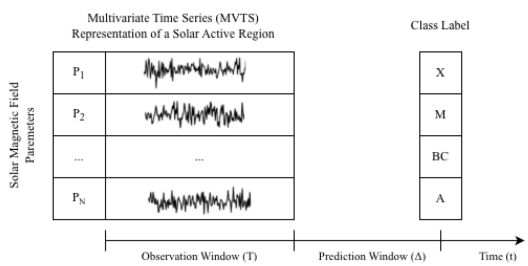


Figure 1: Figure 1: Multivariate time series (MVTS) representation of a solar active region.

At each timestamp t, the magnetic state of an active region is represented by a vector $\mathbf{x}_t \in \mathbb{R}^N$. To capture localized temporal patterns while preserving long-range dependencies, each MVTS instance is partitioned into K equal-length temporal windows. This windowed representation enables joint modeling of short-term dynamics and long-term evolution.

3.2 Dynamic Functional Graph Construction

Magnetic field parameters in solar active regions are not independent but interact through complex physical processes. To explicitly capture these inter-variable dependencies, we represent each temporal window as a weighted functional graph. Nodes correspond to magnetic field parameters, and edges encode statistical similarity between parameter time series within the window.

Edge weights are computed using correlation-based similarity measures, resulting in a graph structure that reflects the strength of inter-variable relationships. Because graphs are constructed independently for each temporal window, the resulting sequence of graphs captures the dynamic evolution of variable interactions over time.

Details of graph construction are provided in Appendix B.

3.3 Graph Attention-Based Spatial Encoding

To learn expressive representations of the constructed functional graphs, we employ a Graph Attention Network (GAT) [Velickovic et al., 2018]. Unlike graph convolutional networks with fixed aggregation schemes, GAT uses self-attention to adaptively weight neighboring nodes, allowing the model to focus on informative inter-variable interactions.

For each temporal window, the GAT processes the node features and graph structure to produce a low-dimensional graph embedding. This embedding captures higher-order spatial dependencies among magnetic field parameters and reflects their relative importance learned directly from data.

3.4 Temporal Modeling with LSTM

The sequence of graph embeddings generated for successive temporal windows forms a high-level temporal representation of the MVTS instance. To model temporal dependencies across windows, we employ a Long Short-Term Memory (LSTM) network.

The LSTM processes the sequence of graph embeddings and captures both short-term transitions and long-range temporal dependencies. The final hidden state serves as a compact representation of the entire MVTS instance, encoding the temporal evolution of inter-variable relationships.

3.5 Model Overview

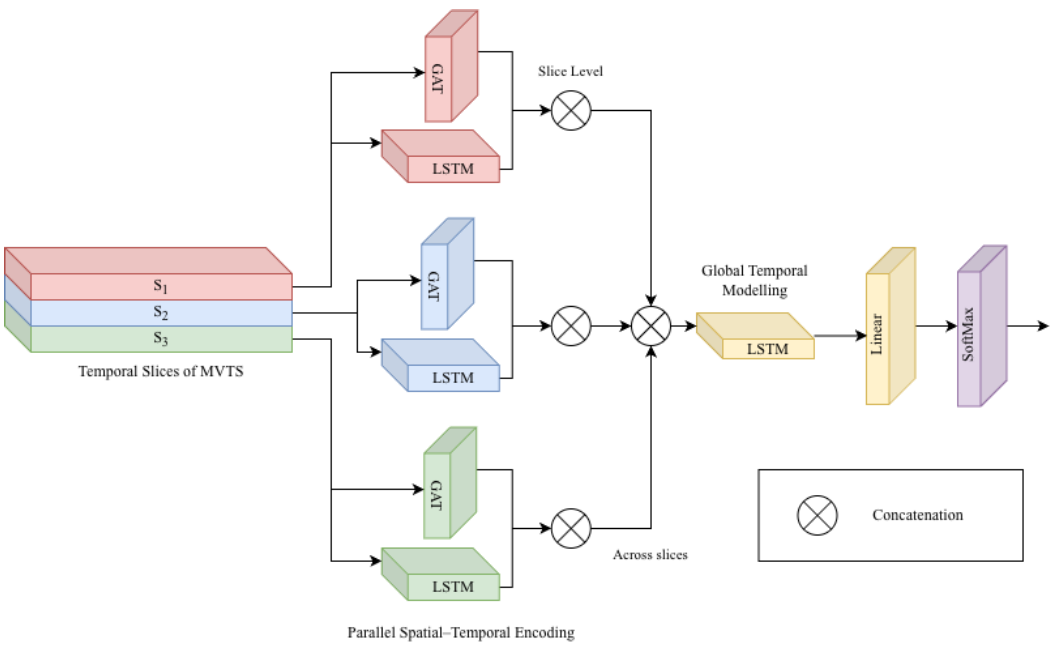


Figure 2: Model Architecture

Figure 2 provides an overview of the SunCast architecture. Given an MVTS instance, the model partitions the input into temporal windows and constructs a functional graph for each window. Graph Attention Networks encode spatial dependencies within each window, producing a sequence of graph embeddings. An LSTM processes this sequence to model temporal dynamics, and the resulting representation is used for flare classification.

3.6 Classification and Training Objective

The final hidden state of the LSTM is passed through a fully connected layer followed by a softmax activation to predict the flare class. The model is trained end-to-end using cross-entropy loss.

All parameters, including graph attention weights and temporal model parameters, are optimized jointly using backpropagation. This end-to-end training enables SunCast to learn both spatial and temporal representations directly from data.

4 Experiments

4.1 Dataset and Experimental Setup

We evaluate SunCast on the multivariate time series (MVTS) solar flare prediction dataset introduced by Angryk et al. [2020] and used in subsequent work [Bahri et al., 2022]. A detailed description of all magnetic parameters is provided in Appendix D.

Each MVTS instance corresponds to a solar active region and consists of a time series of multiple magnetic field parameters observed over a fixed observation window, followed by a prediction window during which flare activity is recorded.

The dataset includes instances labeled according to flare intensity classes, ranging from flare-quiet regions to strong flaring events. We follow the same preprocessing and labeling protocol as prior work to ensure fair comparison. The data is split into training and test sets using a stratified holdout strategy, with two-thirds of the instances used for training and one-third for testing. All reported results are averaged over multiple runs with different random initializations.

Additional training and evaluation details are available in Appendix A.

4.2 Implementation Details

Each MVTS instance is partitioned into equal-length temporal windows. For each window, we construct a functional graph where nodes represent magnetic field parameters and edges are weighted using correlation-based similarity. Graph representations are learned using a Graph Attention Network (GAT), producing a sequence of window-level embeddings.

The sequence of graph-level embeddings is processed by an LSTM to capture temporal dependencies across windows, and the final hidden representation is passed to a linear classifier with softmax activation. All models are trained end-to-end using cross-entropy loss and optimized with the Adam optimizer [Kingma and Ba, 2015].

4.3 Baselines and Evaluation Protocol

We compare SunCast against all baseline methods evaluated in prior work on this dataset. For fair comparison, we re-implemented and re-ran each baseline using the same dataset, preprocessing pipeline, and evaluation protocol.

The baselines include statistical feature-based MVTS classifiers, decision tree–based MVTS models, recurrent neural network–based sequence models, and graph-based models without attention, allowing us to isolate the impact of explicitly modeling inter-variable relationships via attention.

All models are evaluated using standard multiclass classification metrics, including accuracy, precision, recall, and F1-score. Results are reported as mean and standard deviation over multiple runs with different random initializations.

Table 1: Multiclass classification performance comparison on the MVTS solar flare dataset.

Measure	**FLT**	**LTV**	**TS-SUM**	**RNN**	**LSTM**	**ROCKET**	**GCN-LSTM**	**SunCast**
Accuracy	0.259 ± 0.012	0.323 ± 0.020	0.609 ± 0.091	0.427 ± 0.025	0.628 ± 0.030	0.742 ± 0.021	0.817 ± 0.014	$\mathbf{0.88 \pm 0.05}$
Precision (X)	0.232 ± 0.024	0.342 ± 0.041	0.712 ± 0.054	0.534 ± 0.031	0.757 ± 0.028	0.920 ± 0.034	0.932 ± 0.022	$\mathbf{0.96 \pm 0.03}$
Recall (X)	0.264 ± 0.053	0.392 ± 0.043	0.772 ± 0.024	0.631 ± 0.028	0.947 ± 0.023	0.981 ± 0.016	0.990 ± 0.023	$\mathbf{0.98 \pm 0.02}$
F1 (X)	0.244 ± 0.032	0.362 ± 0.040	0.741 ± 0.034	0.582 ± 0.019	0.841 ± 0.014	0.952 ± 0.028	0.961 ± 0.013	$\mathbf{0.97 \pm 0.01}$
Precision (M)	0.254 ± 0.012	0.324 ± 0.033	0.522 ± 0.031	0.411 ± 0.014	0.594 ± 0.018	0.661 ± 0.042	0.803 ± 0.054	$\mathbf{0.85 \pm 0.05}$
Recall (M)	0.260 ± 0.023	0.331 ± 0.061	0.552 ± 0.022	0.402 ± 0.030	0.544 ± 0.014	0.704 ± 0.038	0.824 ± 0.063	$\mathbf{0.90 \pm 0.06}$
F1 (M)	0.257 ± 0.026	0.327 ± 0.042	0.537 ± 0.023	0.406 ± 0.029	0.568 ± 0.020	0.687 ± 0.028	0.811 ± 0.033	$\mathbf{0.87 \pm 0.06}$
Precision (BC)	0.232 ± 0.044	0.263 ± 0.024	0.453 ± 0.033	0.282 ± 0.031	0.495 ± 0.013	0.580 ± 0.026	0.682 ± 0.030	$\mathbf{0.82 \pm 0.07}$
Recall (BC)	0.241 ± 0.053	0.212 ± 0.020	0.472 ± 0.014	0.261 ± 0.021	0.409 ± 0.023	0.573 ± 0.052	0.664 ± 0.050	$\mathbf{0.76 \pm 0.11}$
F1 (BC)	0.236 ± 0.041	0.234 ± 0.024	0.462 ± 0.041	0.271 ± 0.031	0.448 ± 0.031	0.577 ± 0.031	0.673 ± 0.032	$\mathbf{0.79 \pm 0.09}$
Precision (A)	0.324 ± 0.034	0.343 ± 0.044	0.583 ± 0.045	0.483 ± 0.024	0.603 ± 0.024	0.810 ± 0.046	0.831 ± 0.018	$\mathbf{0.87 \pm 0.05}$
Recall (A)	0.251 ± 0.042	0.362 ± 0.071	0.663 ± 0.034	0.413 ± 0.042	0.683 ± 0.023	0.724 ± 0.034	0.772 ± 0.021	$\mathbf{0.86 \pm 0.03}$
F1 (A)	0.282 ± 0.014	0.352 ± 0.013	0.620 ± 0.043	0.445 ± 0.032	0.640 ± 0.024	0.771 ± 0.036	0.798 ± 0.017	$\mathbf{0.87 \pm 0.04}$

NOTE: All baseline results are reproduced under the same experimental setup.

Table 1 summarizes the multiclass classification performance of SunCast and all reproduced baseline methods. SunCast consistently outperforms all baselines across evaluation metrics. Notably, SunCast achieves substantial improvements over both LSTM-based sequence models and non-attentive graph-based models, demonstrating the benefit of attention-based modeling of inter-variable dependencies.

To further illustrate relative performance trends, we visualize accuracy and F1-score across different models using bar plots in figure 3 These visualizations highlight the consistent performance gains achieved by SunCast compared to competing baselines.

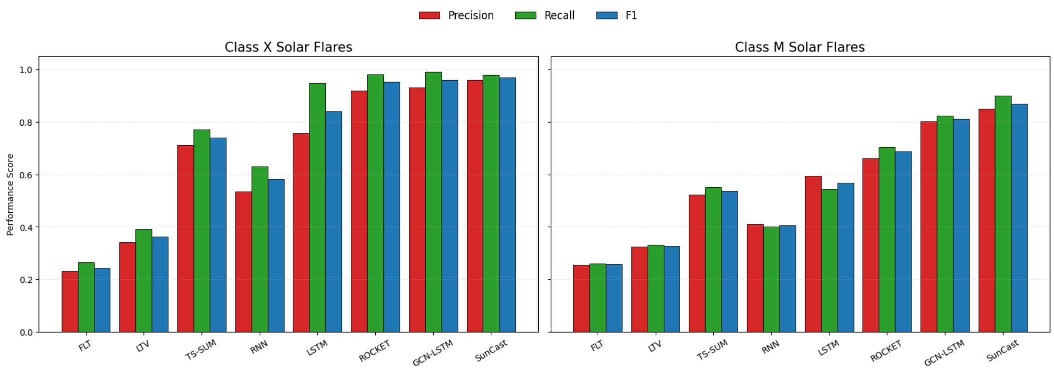


Figure 3: Comparison of results for X and M class Flares

4.4 Binary Flare Prediction Performance

In addition to multiclass prediction, we evaluate SunCast in a binary classification setting, which reflects a common operational use case in solar flare forecasting. Following prior work, X- and M-class events are treated as major flares, while all remaining instances are considered non-flaring events.

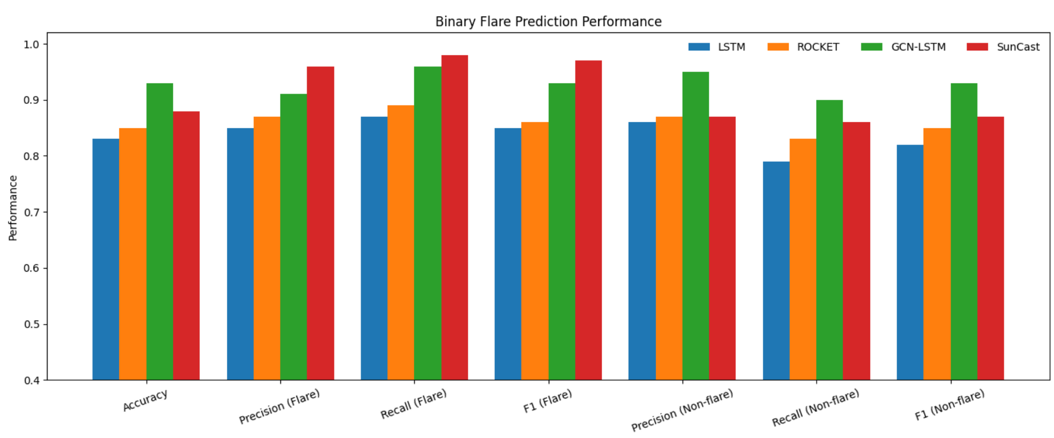


Figure 4: Binary flare prediction performance comparing SunCast with baseline models

Figure 4 reports the binary classification performance in terms of accuracy, precision, recall, and F1-score. SunCast consistently outperforms sequence-only baselines and achieves a higher F1-score compared to non-attentive graph-based models.

4.5 Ablation and Interpretability Analysis

Beyond overall predictive performance, we analyze the internal behavior of SunCast to understand the role of attention-based graph modeling. We conduct ablation studies and visualize learned attention patterns to assess how inter-variable relationships contribute to solar flare prediction.

4.5.1 Attention Pattern Analysis

To examine how SunCast models inter-variable dependencies, we visualize attention weights learned by the graph attention layers. Figure 5 shows attention heatmaps from multiple heads in the second graph attention layer. Each heatmap represents the normalized attention scores between pairs of magnetic field parameters.

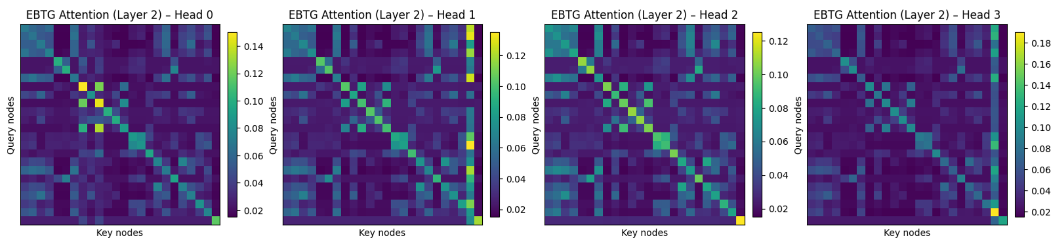


Figure 5: Attention heatmaps from four heads of the second graph attention layer

Distinct attention heads exhibit different interaction patterns, indicating that the model captures complementary relationships among variables. This diversity suggests that multi-head attention enables SunCast to model heterogeneous physical interactions underlying solar magnetic activity.

4.5.2 Attention Head Ablation

To quantify the contribution of individual attention heads, we perform head ablation by selectively disabling one head at a time and measuring the resulting change in model logits. Figure 6 reports the mean change in logit magnitude caused by ablation of each head.

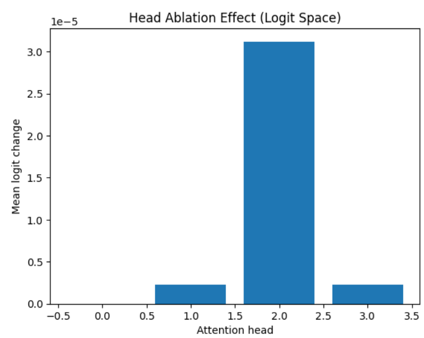


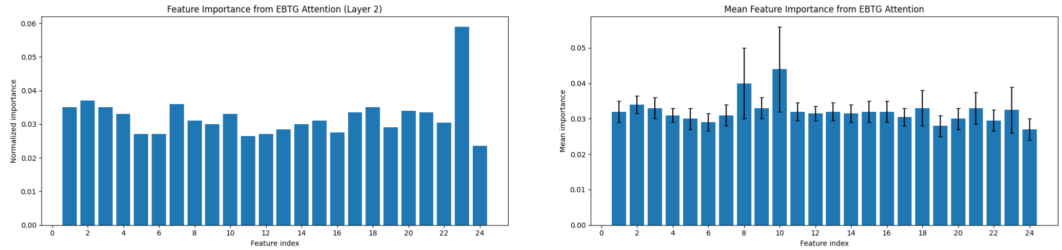
Figure 6: Effect of attention head ablation measured in logit space

The results show that removing certain heads leads to a noticeable degradation in prediction confidence, while others have a smaller effect. This confirms that different attention heads contribute unequally and that the performance gains of SunCast arise from meaningful multi-head attention rather than redundancy.

4.5.3 Feature Importance Analysis

We further analyze feature importance by aggregating attention weights across heads and temporal windows. Figure 7a shows the normalized importance scores of magnetic field parameters for a representative run.

To assess robustness, we compute the mean and standard deviation of feature importance across multiple random initializations. As shown in Figure 7b, the importance rankings remain stable with low variance, indicating consistent and reliable feature attribution. These results suggest that SunCast learns physically meaningful and reproducible inter-variable relationships.



(a) Normalized feature importance scores derived from attention weights for a representative run.

(b) Mean feature importance across multiple runs with standard deviation.

Figure 7: Feature importance analysis based on attention weights.

5 Conclusion and Future Work

We presented *SunCast*, a spatio-temporal deep learning framework for solar flare prediction from multivariate time series (MVTS) data. Unlike prior MVTS-based methods that primarily model temporal dependencies, SunCast explicitly captures inter-variable relationships by representing magnetic field parameters as nodes in attention-based graphs and jointly learning spatial and temporal dynamics.

Experiments on a benchmark solar flare prediction dataset demonstrate that SunCast consistently outperforms strong classical and deep learning baselines in both multiclass and binary settings. Attention visualizations and head ablation studies further provide interpretability, revealing meaningful and stable inter-variable interactions learned by the model.

Future work includes extending SunCast to near–real-time flare forecasting by ingesting continuously updated magnetogram-derived time series from instruments such as the Helioseismic and Magnetic Imager (HMI) aboard the Solar Dynamics Observatory (SDO), enabling rolling-window prediction in operational settings.

Acknowledgements

All authors contributed equally to this work.

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A Training Procedure and Implementation Details

Each multivariate time series (MVTS) instance is divided into fixed-length temporal windows to preserve temporal locality while enabling structured representation learning. For every window, a functional graph is constructed from statistical relationships between magnetic parameters, allowing the model to explicitly capture inter-variable dependencies. Graph embeddings are extracted using a Graph Attention Network (GAT), temporal patterns within windows are captured using a window-level LSTM, and long-range dependencies across windows are modeled using a sequence-level LSTM followed by a linear classifier for final prediction.

This hierarchical design enables simultaneous modeling of spatial parameter interactions and temporal evolution. The complete training procedure is summarized in Algorithm 1.

Algorithm 1 Training of SunCast: GAT–LSTM-based MVTS Embedding

Input: $X_{\text{adj}}, X_{\text{nat}}, y_{\text{train}}, n_{\text{epochs}}, \alpha, \lambda$
Output: Learned parameters $\mathcal{W}$

Initialize $\mathcal{W}$
for epoch $= 1$ to n_{epochs} **do**
    for $i = 1$ to n_{train} **do**
        $Z_w \leftarrow 0$
        for $j = 1$ to η **do**
            $A \leftarrow X_{\text{adj}}[i, j]$
            $X \leftarrow X_{\text{nat}}[i, j]$
            $z_G \leftarrow \mathrm{GAT}(A, X)$
            $z_s \leftarrow \mathrm{LSTM}_s(X^\top)$
            $Z_w[j] \leftarrow \mathrm{Concat}(z_G, z_s)$
        end for
        $z_f \leftarrow \mathrm{LSTM}_f(Z_w)$
        $\hat{y} \leftarrow \mathrm{Softmax}(\mathrm{Linear}(z_f))$
        Update $\mathcal{W}$ using $\mathrm{Adam}(\alpha, \lambda)$
    end for
end for
return $\mathcal{W}$ =0

B Functional Graph Construction

Each window $s \in \mathbb{R}^{\tau \times N}$ is converted into a graph where nodes represent magnetic parameters and edge weights are computed using Pearson correlations between parameter time series. This produces a symmetric fully-connected adjacency matrix that encodes pairwise statistical relationships among variables.

Such correlation-based graphs provide a lightweight yet expressive approximation of physical coupling between parameters. By supplying this structured representation to attention layers, the model can learn adaptive interaction patterns and dynamically emphasize informative relationships while suppressing noisy or weak dependencies.

C Model Architecture

SunCast contains three modules that operate at different representation levels:

- **GAT**: A two-layer multi-head attention network that extracts graph-level embeddings from parameter interaction graphs. Attention allows the model to weight neighboring nodes differently, enabling selective aggregation of informative features.
- **Window LSTM**: Processes node attributes within each window to capture short-term temporal dynamics and local evolution patterns.
- **Sequence LSTM**: Operates on window embeddings to model long-range temporal dependencies and global progression trends across time.

Models are trained using cross-entropy loss and Adam optimization. Hyperparameters are selected through validation experiments and kept fixed across runs to ensure fairness and reproducibility.

D SHARP Dataset Parameter Description

Table 2 lists the SHARP magnetic parameters used as node variables in the functional graphs.

Table 2: Magnetic field parameters used in experiments

Feature	Parameter	Description
1	ABSNJZH	Absolute net current helicity
2	EPSX	Sum of X-component of normalized Lorentz force
3	EPSY	Sum of Y-component of normalized Lorentz force
4	EPSZ	Sum of Z-component of normalized Lorentz force
5	MEANALP	Mean twist parameter
6	MEANGAM	Mean inclination angle
7	MEANGBH	Mean horizontal field gradient
8	MEANGBT	Mean total field gradient
9	MEANGBZ	Mean vertical field gradient
10	MEANJZD	Mean vertical current density
11	MEANJZH	Mean current helicity
12	MEANPOT	Mean excess magnetic energy density
13	MEANSHR	Mean shear angle
14	R_VALUE	Unsigned flux near polarity inversion lines
15	SAVNCPP	Sum absolute net current per polarity
16	SHRGT45	Area with shear angle $> 45°$
17	TOTBSQ	Total magnitude of Lorentz force
18	TOTFX	Sum X-component Lorentz force
19	TOTFY	Sum Y-component Lorentz force
20	TOTFZ	Sum Z-component Lorentz force
21	TOTPOT	Total magnetic energy density
22	TOTUSJH	Total unsigned current helicity
23	TOTUSJZ	Total unsigned vertical current
24	USFLUX	Total unsigned magnetic flux

E Evaluation Protocol

Data is split using stratified hold-out (2:1 train/test). Experiments are repeated across multiple random seeds and results reported as mean $\pm$ standard deviation.

Metrics include accuracy, precision, recall, and F1-score. For binary tasks, X/M flares are treated as positive events and others as negative.

F Interpretability Summary

To better understand learned representations, attention weights from the final GAT layer are analyzed to identify dominant parameter interactions. These attention patterns reveal which variable relationships consistently influence predictions across samples.

Head ablation experiments further show that only a subset of attention heads contributes strongly to performance, suggesting functional specialization among heads. Aggregated attention scores are used to estimate feature importance, providing interpretable insights into which magnetic parameters most strongly affect model decisions.